

Curried Functional Spaces

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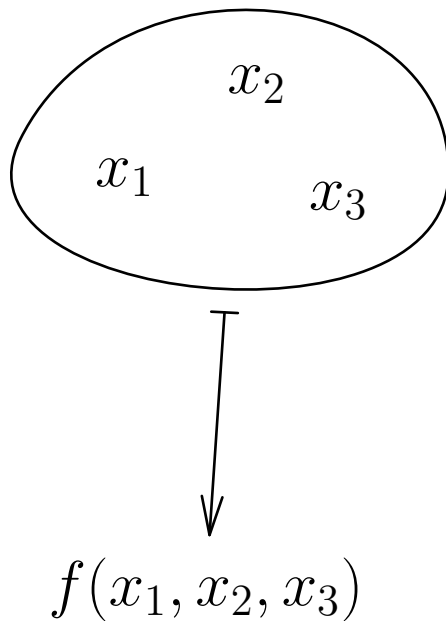
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Premise

Multivariable functions

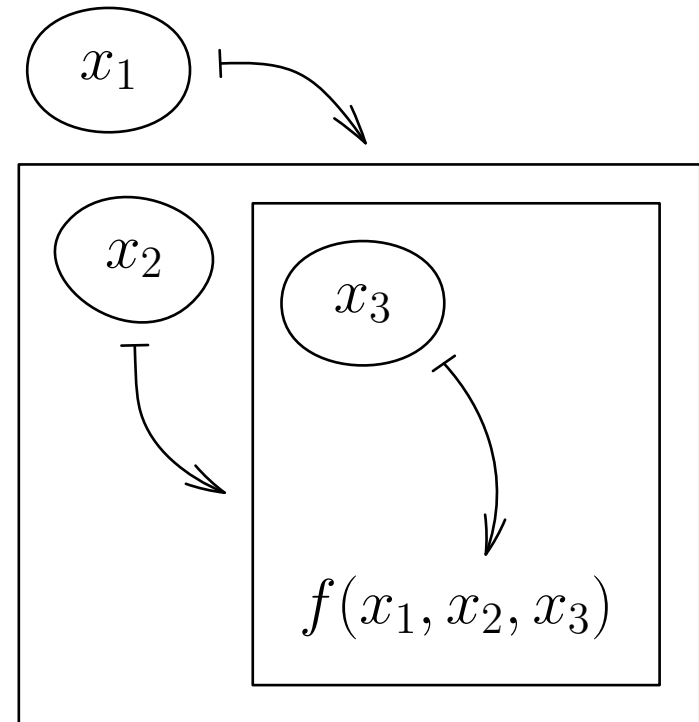
$$\mathbb{R}^n \rightarrow \mathbb{R}$$



Multivariate calculus

Curried functions

$$\mathbb{R} \rightarrow (\mathbb{R} \rightarrow (\dots (\mathbb{R} \rightarrow \mathbb{R})))$$

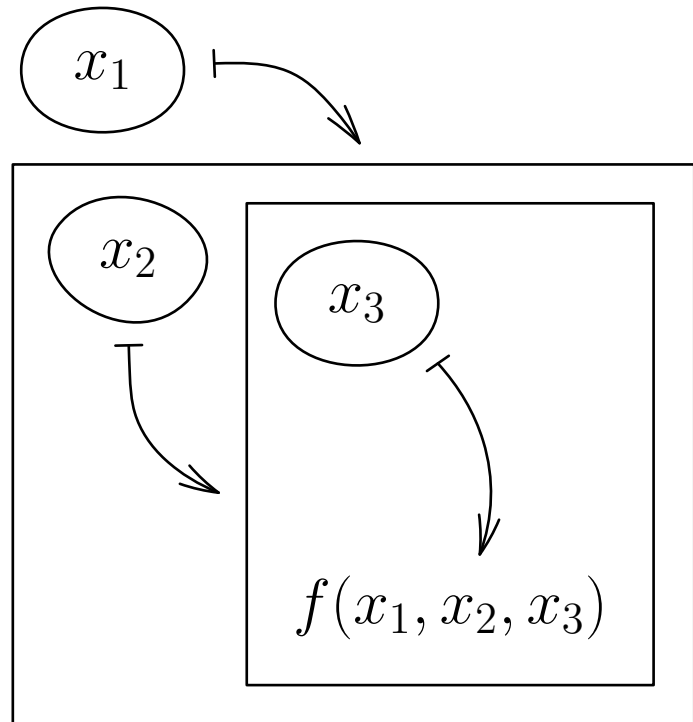


λ -calculus, functional programming

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Curried functions

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λ -calculus,
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Multivariate calculus

Continuous Functions

$C_n(\mathbb{R})$ — the space of continuous n -ary curried functions

1. $C_0(\mathbb{R}) := \mathbb{R}$

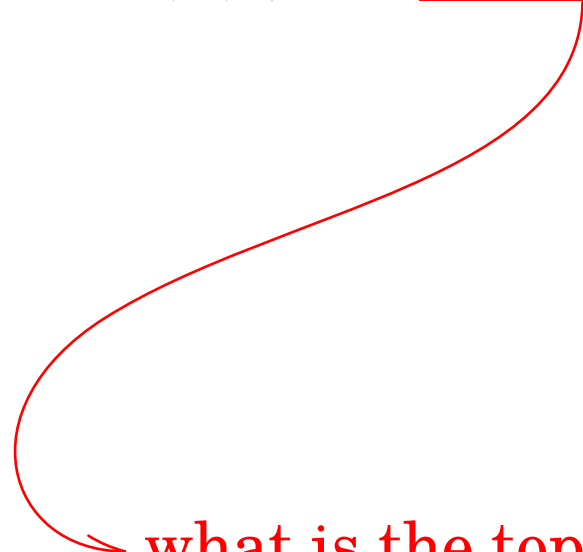
2. $C_{n+1}(\mathbb{R}) := \{f: \mathbb{R} \rightarrow C_n(\mathbb{R}) \mid f \text{ is continuous}\}$

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⇒ what is the topology on $C_n(\mathbb{R})$?

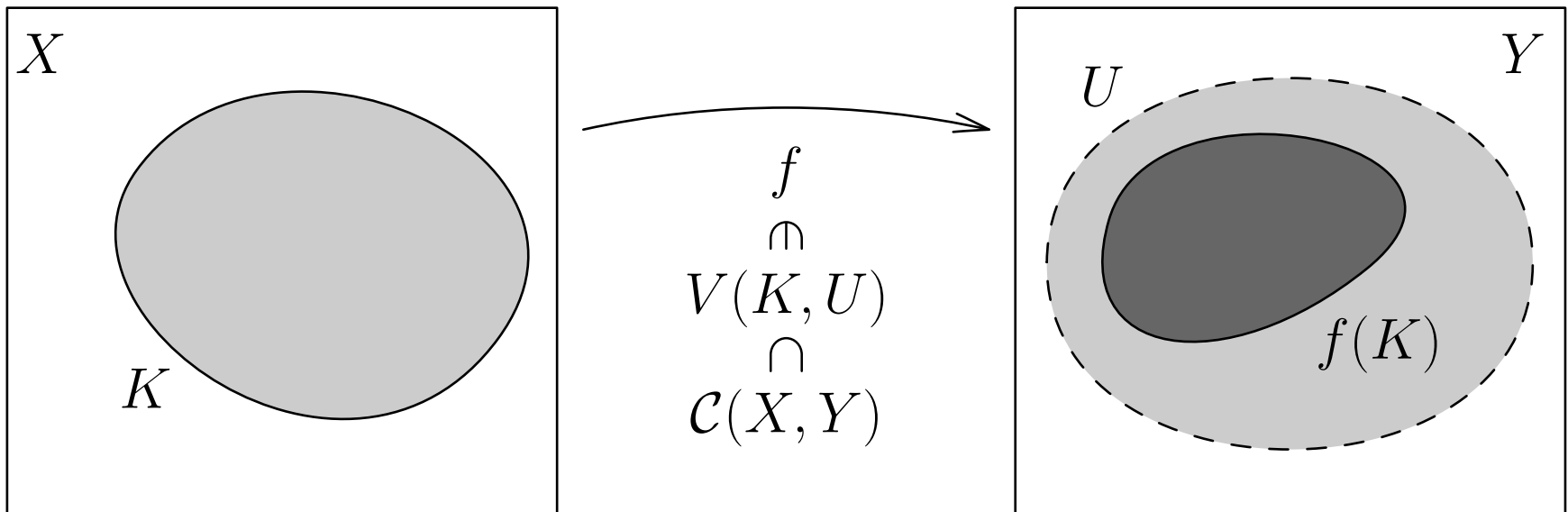
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↪ compact-open topology: $C_n(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_{n-1}(\mathbb{R}))$



Topological Vector Spaces

Remark: If $f_1, f_2 \in C_n(\mathbb{R})$, then $\alpha f_1 + \beta f_2 \in C_n(\mathbb{R})$

Definition: X is a **TVS** $\Leftrightarrow X$ has **topological** and **vector** structure, and they are *compatible*:

1. $(+): X \times X \rightarrow X$ is continuous;
2. $(\cdot): \mathbb{R} \times X \rightarrow X$ is continuous.

